

10.6 Appendix

10.6.1 Appendix 10.A: Interpretation of MSK as OQPSK

In this appendix, we show that MSK can be interpreted as OQPSK. Consider the real part of an MSK signal over the duration $2T_B$:

$$\begin{aligned}
 & (k)T_B \leq t < (k+2) \cdot T_B \text{ with } k \text{ even} \\
 \text{Re}\{s(t)\} &= \sqrt{2E_B/T_B} \cos \left\{ \omega_{\text{mod}} \sum_{i=-\infty}^{k-1} b_i T_B + b_k \omega_{\text{mod}} \int_{kT_B}^t p_{\text{D,MSK}}(\tau) d\tau \right\} \\
 &= \sqrt{2E_B/T_B} \cos \left\{ \varphi_{\text{prev}}(k-1) + b_k \omega_{\text{mod}} \int_{kT_B}^t p_{\text{D,MSK}}(\tau) d\tau \right\} \\
 &= \sqrt{2E_B/T_B} \left(\cos[\varphi_{\text{prev}}(k-1)] \cos \left[b_k \omega_{\text{mod}} \int_{kT_B}^t p_{\text{D,MSK}}(\tau) d\tau \right] \right. \\
 &\quad \left. - \sin[\varphi_{\text{prev}}(k-1)] \sin \left[b_k \omega_{\text{mod}} \int_{kT_B}^t p_{\text{D,MSK}}(\tau) d\tau \right] \right) \tag{10.90}
 \end{aligned}$$

where $\varphi_{\text{prev}}(k) = \omega_{\text{mod}} \sum_{l=-\infty}^{k-1} b_l T_B$ denotes the phase at the beginning of the observation time and $\omega_{\text{mod}} = 2\pi f_{\text{mod}}$. Depending on the previous bits, that phase is

$$(\varphi_{\text{prev}}(k-1)) \bmod(2\pi) = \begin{cases} 0 \text{ or } \pi \\ \frac{\pi}{2} \text{ or } \frac{3 \cdot \pi}{2} \end{cases} \tag{10.91}$$

Without restriction of generality, we consider the second case (starting phase equal to $\pi/2$ or $3\pi/2$). Thus

$$\begin{aligned}
 \cos(\varphi_{\text{prev}}(k-1)) &= 0 \\
 \sin(\varphi_{\text{prev}}(k-1)) &= \pm 1 \tag{10.92}
 \end{aligned}$$

and

$$\begin{aligned}
 \text{Re}\{s(t)\} &= \sqrt{2E_B/T_B} \sin[-\varphi_{\text{prev}}(k-1)] \sin \left[b_k \omega_{\text{mod}} \int_{kT_B}^t p_{\text{D,MSK}}(\tau) d\tau \right] \\
 &= \sqrt{2E_B/T_B} \sin[-\varphi_{\text{prev}}(k-1)] b_k \sin[\omega_{\text{mod}}(t - kT_B)] \tag{10.93}
 \end{aligned}$$

For the imaginary part, we consider a time interval of length $2T_B$ that is offset by T_B with

respect to the observation interval of the real part:

$$\begin{aligned}
& (k+1)T_B \leq t < (k+3)T_B \\
\text{Im}\{s(t)\} &= \sqrt{2E_B/T_B} \sin \left\{ \omega_{\text{mod}} \sum_{l=-\infty}^k b_l T_B + b_k \omega_{\text{mod}} \int_{(k+1)T_B}^t p_{\text{D,MSK}}(\tau) d\tau \right\} \\
&= \sqrt{2E_B/T_B} \cos \left\{ \varphi_{\text{prev}}(k) + b_k \omega_{\text{mod}} \int_{(k+1)T_B}^t p_{\text{D,MSK}}(\tau) d\tau \right\} \\
&= \sqrt{2E_B/T_B} \left(\cos[\varphi_{\text{prev}}(k)] \sin \left[b_k \omega_{\text{mod}} \int_{(k+1)T_B}^t p_{\text{D,MSK}}(\tau) d\tau \right] \right. \\
&\quad \left. + \sin[\varphi_{\text{prev}}(k)] \cos \left[b_k \omega_{\text{mod}} \int_{(k+1)T_B}^t p_{\text{D,MSK}}(\tau) d\tau \right] \right) \tag{10.94}
\end{aligned}$$

From $\varphi_{\text{prev}}(k-1) = \pi/2$ we find $\varphi_{\text{prev}}(k) = 0/\pi$ and $\cos(\varphi_{\text{prev}}(k)) = \pm 1$ and $\sin(\varphi_{\text{prev}}(k)) = 0$, respectively

$$\begin{aligned}
\text{Im}\{s(t)\} &= \sqrt{2E_B/T_B} \cos[\varphi_{\text{prev}}(k)] \sin \left[b_k \omega_{\text{mod}} \int_{(k+1)T_B}^t p_{\text{D,MSK}}(\tau) d\tau \right] \\
&= \sqrt{2E_B/T_B} \cos[\varphi_{\text{prev}}(k)] b_{k+1} \sin[\omega_{\text{mod}}(t - (k+1)T_B)] \tag{10.95}
\end{aligned}$$

This derivation shows that MSK can be interpreted as OQAM, where the impulse shape is a half-sinewave of duration $2T_B$. The sign of the pulse is determined by both the “history” (i.e., the starting phase) and the bit of the current observation symbol. This is shown in Fig. 10.35.